Land cover



Vegetation · NDVI

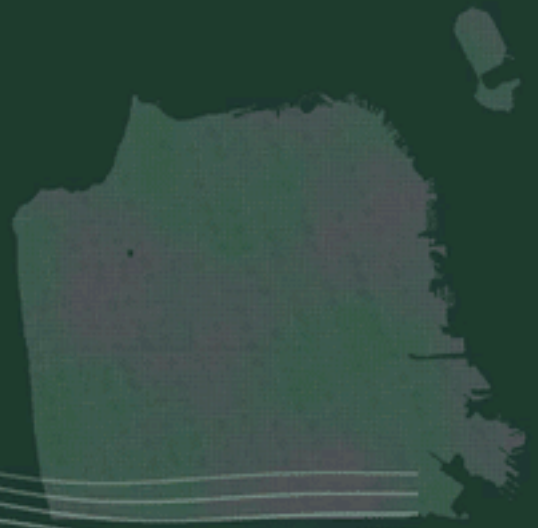


Soil carbon



# planIt

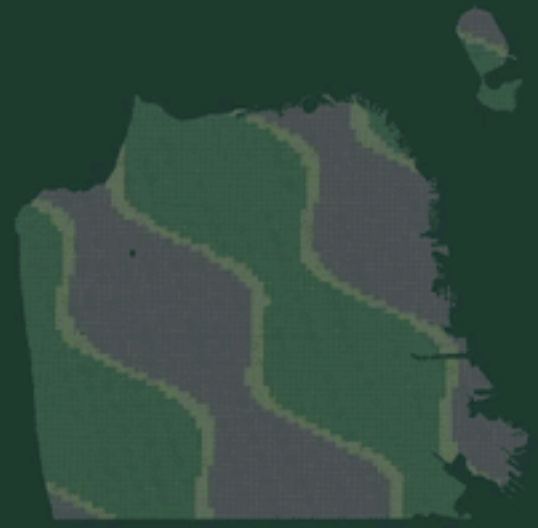
Productivity



Carbon change



Land degradation



## Richard Rex

ML & Earth Science

Carbon sequestration

Hydrogen storage

## Kiran Kashyap

Business & Communication

Consumer products

Consulting

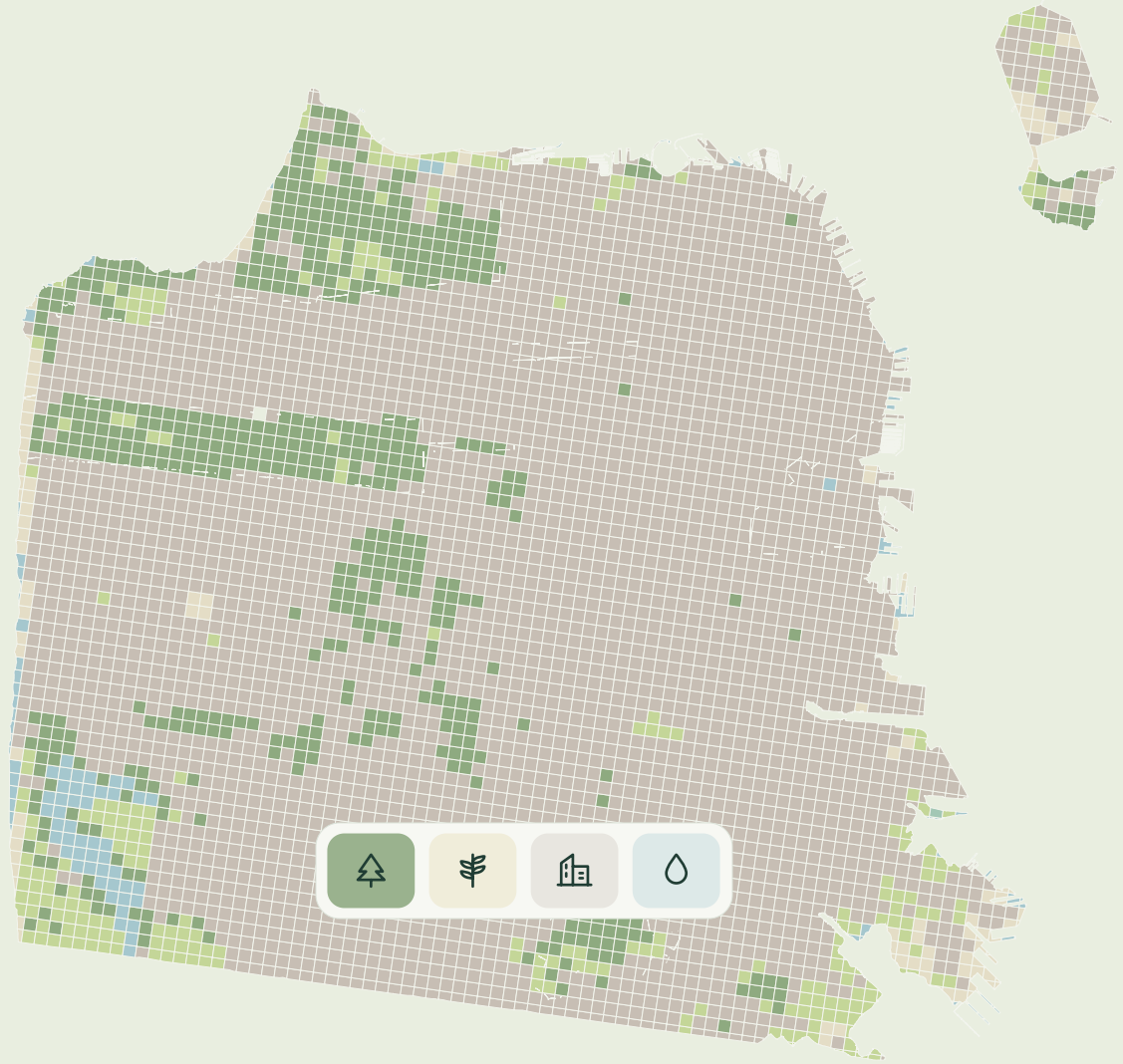
VC

SDG 15 · LIFE ON LAND

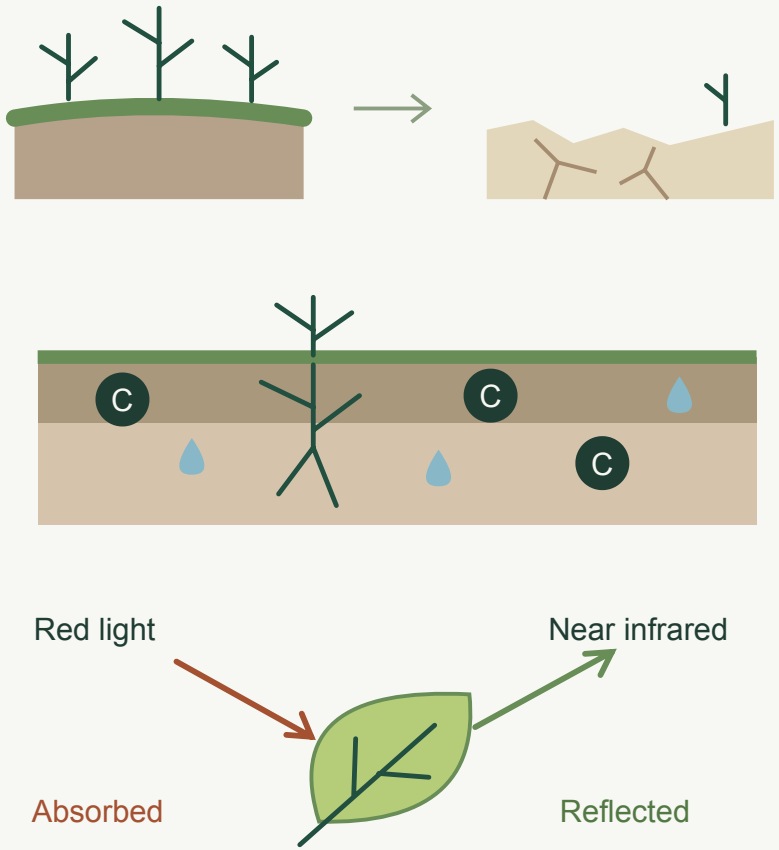
"Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss"

KEY INDICATOR · 15.3.1

"proportion of land that is degraded over total land area"



# What are we measuring?




## What is degradation?

A lasting loss of the land's ability to support life and livelihoods.

## Why soil carbon?

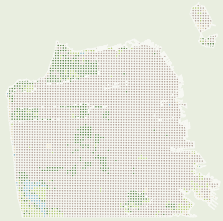
Organic matter supports water and nutrient storage. Its carbon content helps indicate soil condition.

## What does NDVI measure?

Vegetation greenness, measured from reflected light. A proxy for vegetation activity.

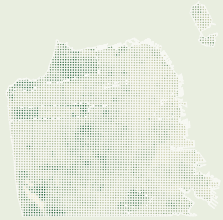
SDG 15.3.1 · LAND DEGRADATION

# Signals of land degradation



## Land cover

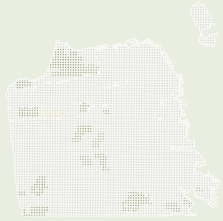
Compare land-cover classes



## Vegetation index

NDVI over time → productivity

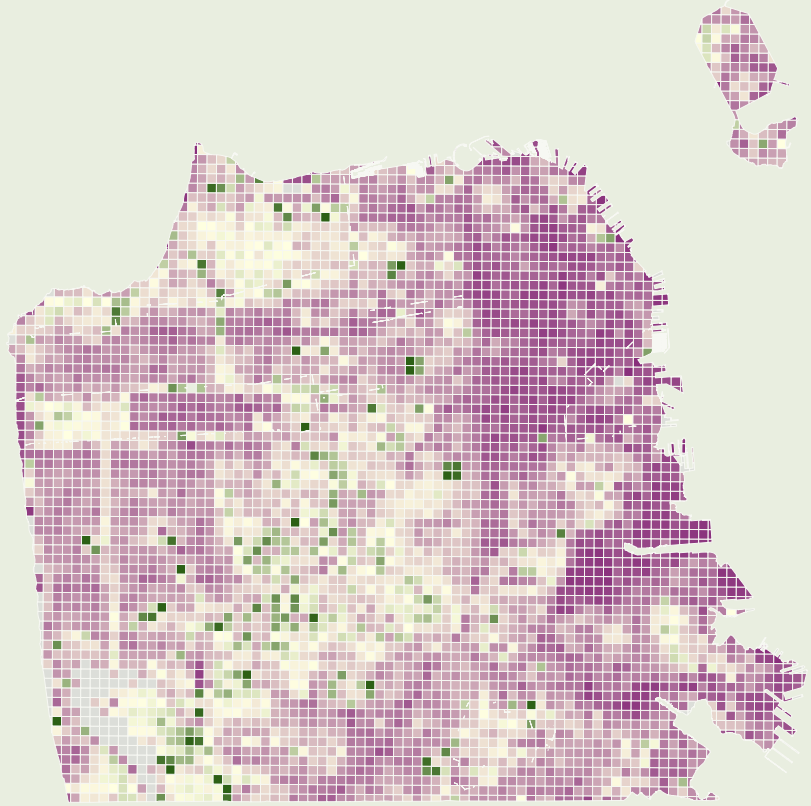
≤ −0.1 ≥ 0.8



## Soil carbon

Compare carbon stocks · t C/ha

0 ≥ 120



## Land degradation

Productivity performance · NDVI / benchmark

≤ 0 1 ≥ 1.5

# Design considerations

Create an engaging experience for ages 15–24
to explore land-use choices, understand their consequences,
and participate in local planning.

# Block by Block

"Minecraft lowers the barrier so that anyone could actually have a say in their public space."

Lydia Winters · Mojang

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## 55+ countries

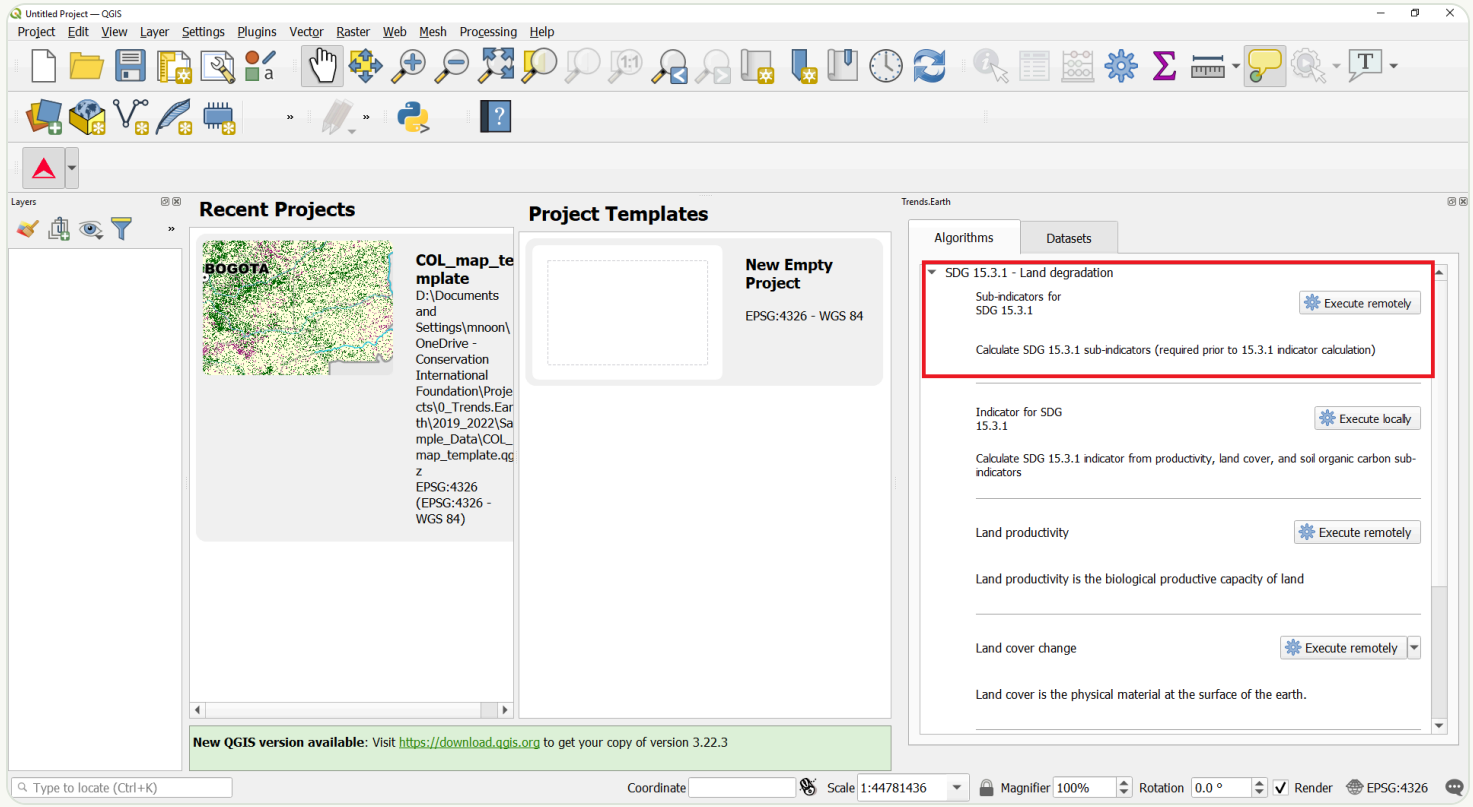
Projects funded or activated by the foundation

People set priorities.
Agents help compare.



Young people presenting ideas · Hanoi, Vietnam
Photo: Microsoft team / Block by Block

# Trends.Earth: from data to assessment.




## An expert GIS workflow

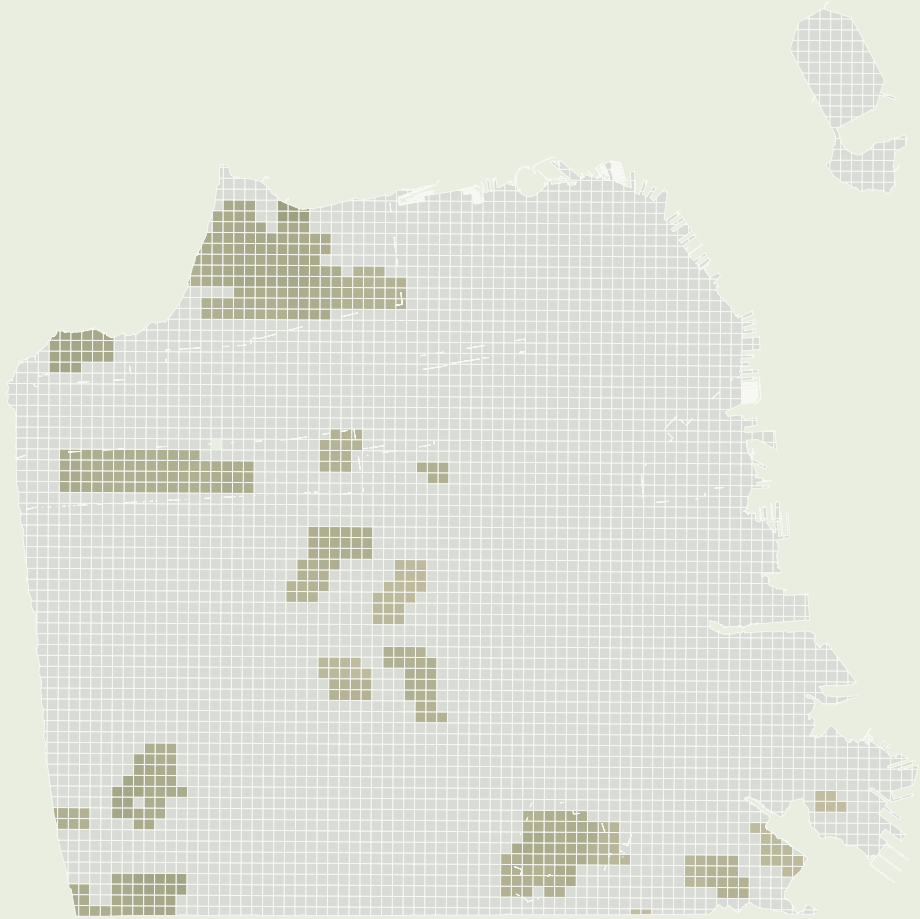
01 Select an area and time period

02 Calculate three land-condition indicators

03 Review maps and area totals

planIt: paint a change → simulate → compare.

# Real data has gaps.

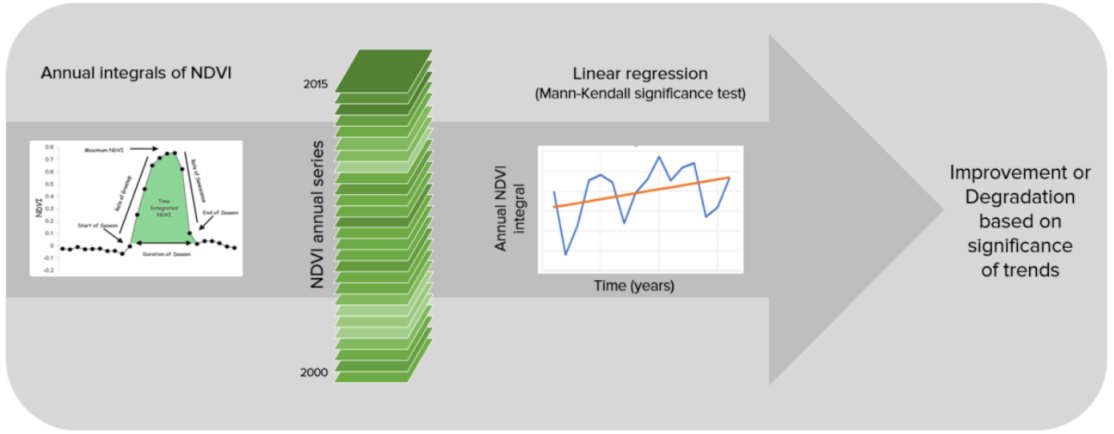


**93%**

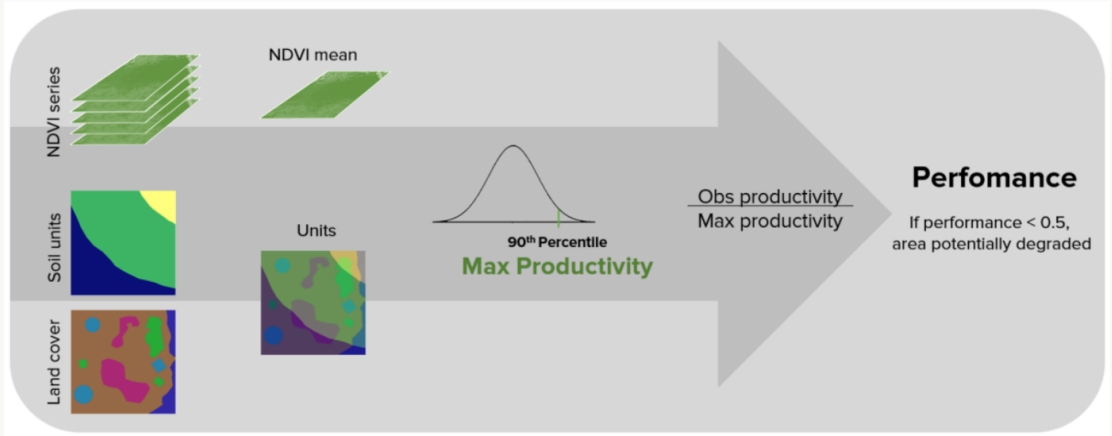
of this SF study area lacks usable soil-carbon values.

# Complex modeling

## Trends.Earth · trajectory




## Trends.Earth · performance




## Satellite measurement

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}}$$

Red and near-infrared reflectance

## Our scenario choice

$$N(t) = N_0 + (N_* - N_0)\min\left(\frac{t}{\tau}, 1\right)$$

Tree response: τ = 10 years, assumed

# Demo

# Under the hood

Simulation, model assumptions and agent tools.

# Modeling assumptions

## Soil carbon

$$C^* = C_e \sum_k f_{k,e} F_{k \to j}(\text{climate})$$

$$C(t) = C_e + (C^* - C_e)\ \text{clamp}\left(\frac{t - t_e}{20}, 0, 1\right)$$

$$\Delta C = 100\left(\frac{C(t)}{C_0} - 1\right), \qquad |\Delta C| \geq 10\%$$

≤ −10% declining · ≥ +10% improving

Top 30 cm · t C/ha · missing or nonpositive $C_0$: unknown

## Vegetation · our scenario assumption

$$N^* = \text{clamp}\left(N_0 + \sum_k \beta_k (f_{k,\text{new}} - f_{k,0}), -1, 1\right)$$

$$N(t) = N_e + (N^* - N_e)\ \text{clamp}\left(\frac{t - t_e}{\tau_j}, 0, 1\right)$$

τ = 0: replace the surface immediately.

Later edits restart from the attained value.

### Carbon conversion F · row → column

| | Tree | Grass | Crop | Wet | Built | Bare | Water |
|---|---|---|---|---|---|---|---|
| **Tree / grass** | 1 | 1 | f | 1 | .1 | .1 | 1 |
| **Crop** | 1/f | 1/f | 1 | 1/.71 | .1 | .1 | 1 |
| **Wetland** | 1 | 1 | .71 | 1 | .1 | .1 | 1 |
| **Built / bare** | 2 | 2 | 2 | 2 | 1 | 1 | 1 |
| **Water** | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

f: temperate dry .80 / moist .69

tropical dry .58 / moist .48 / montane .64

### Vegetation coefficients · β / response years τ

| Tree | Grass | Crop | Wet | Built | Bare | Water |
|---|---|---|---|---|---|---|
| .67 | .64 | .55 | .65 | .10 | .10 | −.10 |
| 10 yr | 2 yr | 1 yr | 3 yr | 0 yr | 0 yr | 0 yr |

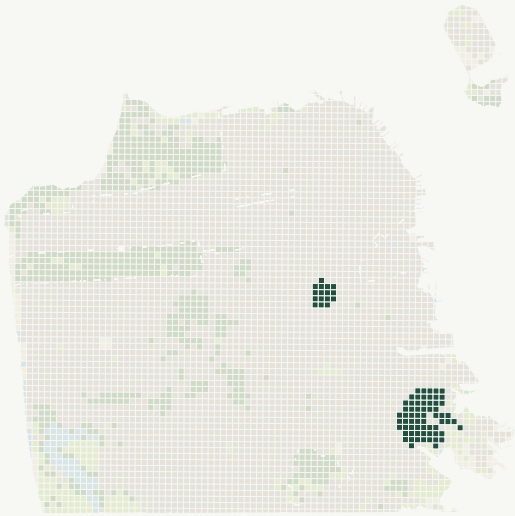
Tree / grass / built β: SF observational fit.

Other β: placeholders. All durations: assumptions.

Documented coefficients ≠ locally validated forecasts.

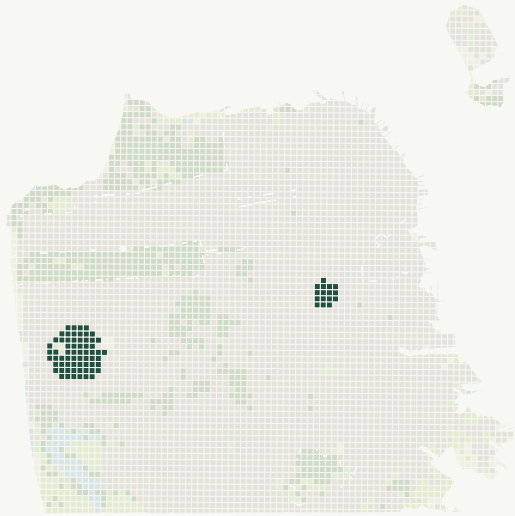
# Exploring land-use alternatives

EXAMPLE REQUEST "Try the same area of greening in three different places."



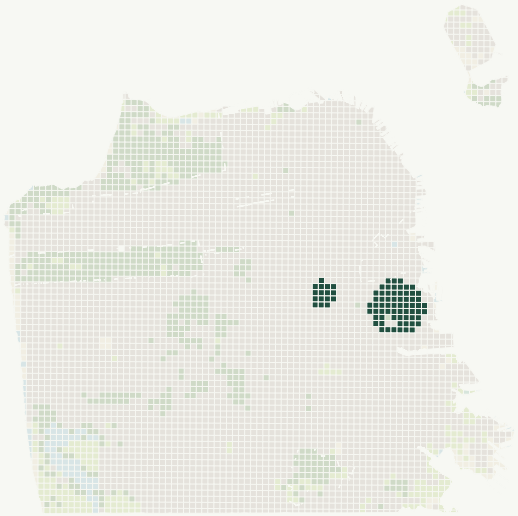
**Bayview**

Inspect real cells



**Sunset / Parkside**

→ Duplicate the starting design



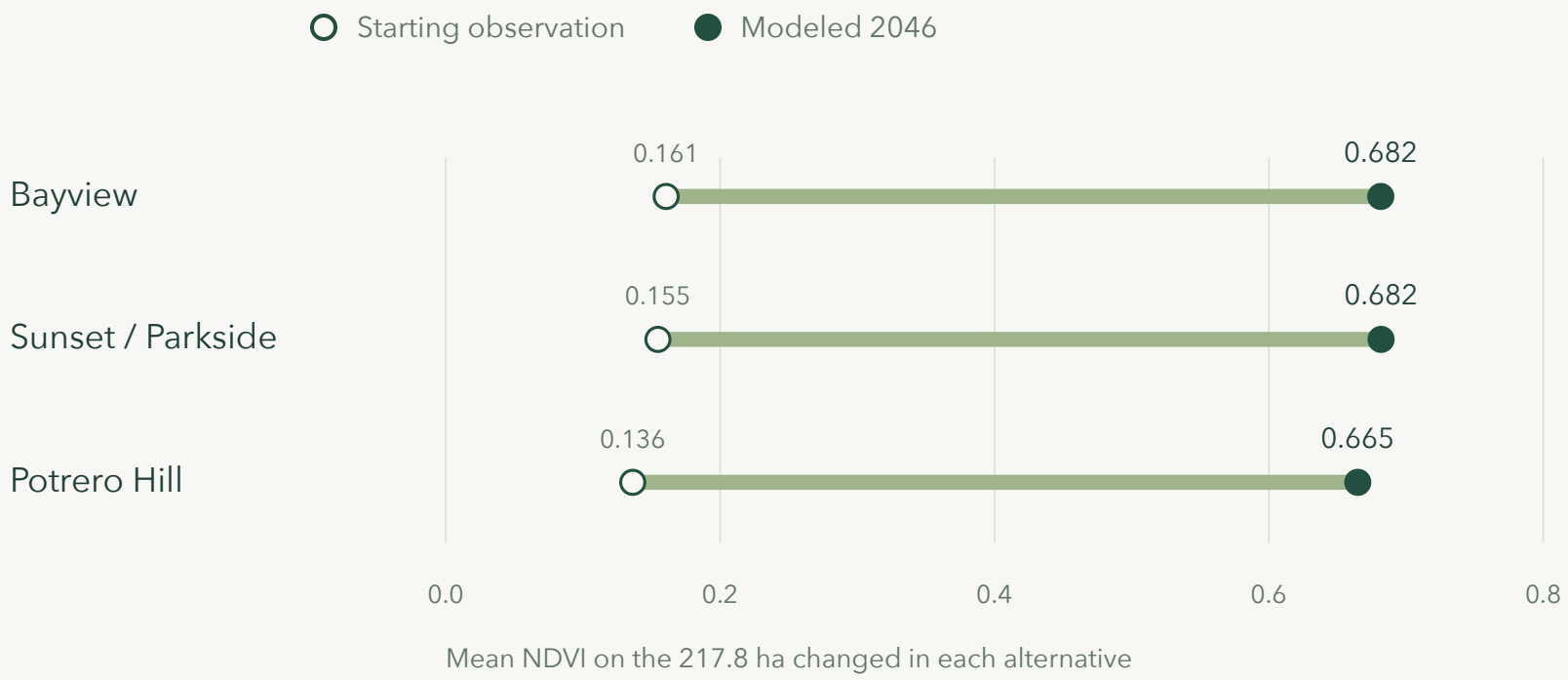
**Potrero Hill**

→ Save dated edits

**80 cells each · 217.8 ha** 16 shared Mission cells + 64 alternative cells · whole-cell tree substitutions

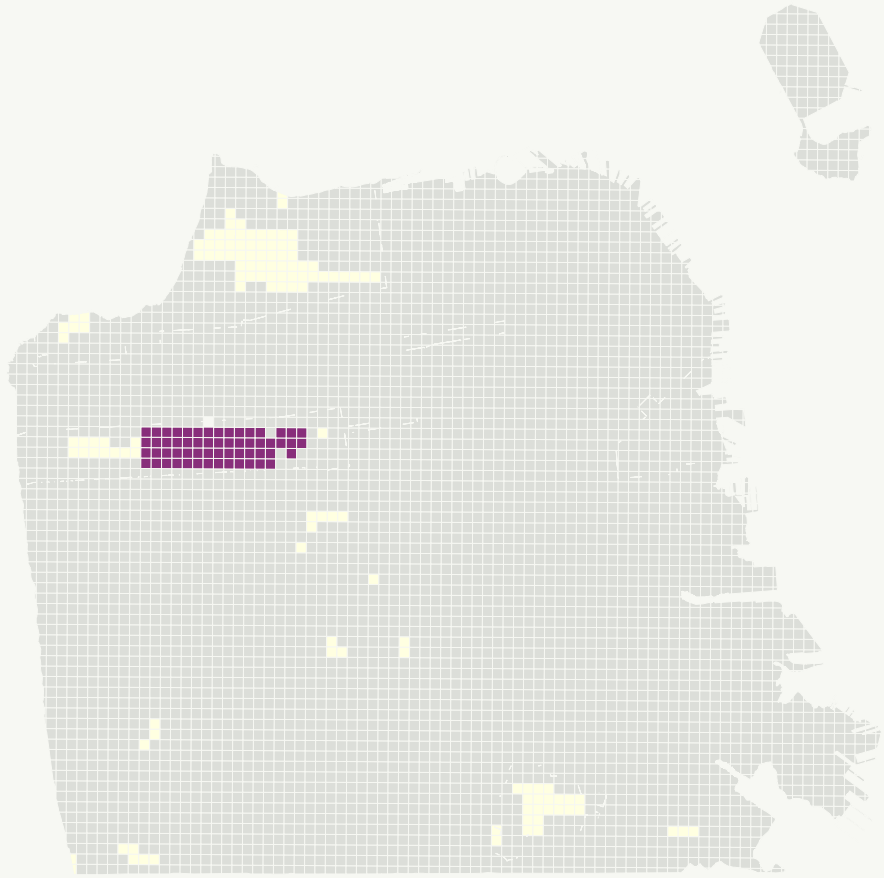
# Comparing modeled NDVI

EXAMPLE REQUEST "Compare the three designs at 2046."




Mean NDVI on the 217.8 ha changed in each alternative

# “What % of my land is degrading?”



SF tree → built scenario · 2046

COMPUTED SCENARIO RESULT

**1.3%**

of total land is flagged declining

$$\frac{157.9\ \text{ha}}{12{,}199.4\ \text{ha}} \times 100$$



Declining  Stable  Unassessed

92.5% remains unassessed.
The rest is not automatically healthy.

# Three-month development plan

## Game mechanics

### Scenario objectives

Restoration, farming and development

### Resource constraints

Budgets, land and competing needs

### Collaborative planning

Propose, discuss and revise together

## Resolution & data

### Finer decision grids

More detail where source data supports it

### Terrain, rainfall and soil

Add inputs beyond the current baseline

### Additional hackathon resources

FAO Earth Map + NatCap Data Hub

## Ecological simulation

### Water and soil movement

Runoff, infiltration and erosion

### Interactions between cells

Upstream changes, downstream effects

### Ecosystem-service models

Integrate a relevant InVEST model

# Data sources and methods

## Observations

ESA WorldCover ↗

Land cover · 2021 v200

Sentinel-2 composites ↗

Observed vegetation index

ISRIC SoilGrids ↗

Soil organic carbon · top 30 cm

## Methods & assumptions

Trends.Earth methods ↗

Assessment logic & SOC coefficients

model.py · ndvi.py

Versioned scenario implementation

post-demo-provenance.json

Inputs, selected cells, calculated values

## Purpose & participation

UN SDG 15 ↗

Restoration & land degradation

Block by Block ↗

Community-led public-space design

Our next evidence

Learning impact remains to be measured